Data-Backed Journal-Style Replication of *Exploring Uncharted Territory*

Replication Paper Project

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Abstract

This paper presents a local calculated replication of Exploring Uncharted Territory: Knowledge Search Processes in the Origination of Outlier Innovation using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The current local batch contains 23,901 observations across 184 entities, and the batch headline estimate is 0.0203 for Distant Recombination with p = 0.020. The local analogue finds positive and statistically meaningful coefficients on long search paths and distant recombination, but not on the firm-level science signal used to proxy scientific reasoning. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: outlier innovation; long search paths; scientific reasoning; distant recombination; local calculated analogue

1 Introduction

This replication begins from a paper-first reading of Exploring Uncharted Territory: Knowledge Search Processes in the Origination of Outlier Innovation and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The motivating question is: The source paper asks which search processes are associated with the origination of outlier innovation

(Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The study is worth revisiting because The replication matters because the paper is a rare strategy study that makes a strong process claim about how inventors reach territory far from what is already known. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The executable evidence base used in this manuscript contains 23,901 observations across 184 entities and spans 1986 to 2016 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (Kneeland et al., 2020; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper's contribution is a transparent processed-dataset analogue that preserves the three-process logic while making the outlier and science proxies explicit. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (Kneeland et al., 2020; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper belongs to a top-journal conversation about exploratory search, inventive distance, knowledge recombination, and the search processes that generate unusually important innovation. (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Cohen and Levinthal, 1990; Kogut and Zander, 1992, 1996; Teece et al., 1997; Benner and Tushman, 2003). That literature matters because the replication is explicitly about process recovery: it asks whether the three pathways proposed in the source article remain visible when the local rerun works from the processed data stack already available in the workspace. (Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Henderson and Clark, 1990; Anderson and Tushman, 1986; Hargadon and Sutton, 1997; Tripsas and Gavetti, 2000; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Stuart, 2000). The present manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration, recombination, performance measurement, alliance-based learning,

inventive search, technological discontinuity, and patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference base rather than on the source article alone. (Rosenkopf and Almeida, 2003; Schilling and Phelps, 2007; Almeida and Kogut, 1999; Fleming et al., 2007; Jaffe, 1986; Jaffe et al., 1993; Trajtenberg, 1990; Hall et al., 2005; Henderson and Cockburn, 1994). The paper-specific review supplements the standard exploration literature with top-journal work on scientific search, recombinant uncertainty, and breakthrough discovery so the outlier-innovation mechanism is grounded in a broader intellectual context. (Fleming and Sorenson, 2004; Nerkar, 2003). A second paper-specific literature block is added so that the manuscript's topic-matched references are carried into the final bibliography explicitly rather than being silently implied by the broader shared review. (Lavie et al., 2011; Ghosh et al., 2014). A second paper-specific literature block is added so that the manuscript's topic-matched references are carried into the final bibliography explicitly rather than being silently implied by the broader shared review. (Rivkin and Siggelkow, 2003).

2 Original Study, Theory, and Replication Boundary

The focal source article is (Kneeland et al., 2020). The original study can be summarized as a mechanism paper with a sharp empirical core. The published paper proposes long search paths, scientific reasoning, and distant recombination as the three main origination processes. Outlier innovation rather than ordinary innovation is the focal outcome. The source study combines large-scale data with close qualitative interpretation. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (Kneeland

et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

2.1 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Three processes are central: long search paths, scientific reasoning, and distant recombination. The local reconstruction therefore adopts the following bounded decision: The local rerun maps those three processes to patent-level search depth, firm-year science exposure, and recombination breadth. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Outlier innovation is the focal outcome. The local reconstruction therefore adopts the following bounded decision: Outlier status is proxied with the local top-tail impact indicator. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The source design uses richer qualitative evidence than a pure regression paper. The local reconstruction therefore adopts the following bounded decision: The paper is written as a transparent analogue because inventor-level interview evidence is not part of the current local

dataset stack. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

DirectSourceFact	LocalReplicationDecision
Three processes are central: long search paths, scientific reasoning, and distant recombination.	The local rerun maps those three processes to patent-level search depth, firm-year science exposure, and recombination breadth.
Outlier innovation is the focal outcome.	Outlier status is proxied with the local top-tail impact indicator.
The source design uses richer qualitative evidence than a pure regression paper.	The paper is written as a transparent analogue because inventor-level interview evidence is not part of the current local dataset stack.

Direct source facts and explicit local execution decisions carried in the body of the paper.

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Processed patent-level recombination panel from the local cleaned reproduction dataset. Science-linked firm-year panel built from publication stocks and NPL-related counts in the current workspace. Outlier outcome drawn from the local top-tail patent-impact indicator (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). Operationally, the reconstruction relies on four evidence layers: the analysis

panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 23,901 observations across 184 entities and covers 1986 to 2016 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The local analogue maps the three proposed processes onto a processed patent panel: long search paths are proxied by prior-art search depth, scientific reasoning by science-linked firm-year exposure, and distant recombination by recombination breadth. The local sample frame is Patent-level recombination panel with science-linked firm-year exposure merged in by firm and application year The observed time window in the executable panel is 1986 to 2016 in the local executable panel The outcome is Top-tail outlier indicator from the local yearly patent-impact distribution (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (Kneeland et al., 2020; Hall et al., 2005).

3.1 Executable File Selection and Data Availability

The present manuscript works from the processed datasets and calculated exports already available in the workspace for Kneeland, Schilling, and Aharonson (2020). That does not mean the paper skips data analysis. It means the data analysis begins from the current executable panel, current coefficient export, and current sample-summary files rather than from a freshly rebuilt raw archive. The substantive obligation is therefore to explain exactly why each verified file is needed and how it enters the paper's argument (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). The analysis panel used in this paper serves a specific empirical role: it is needed to estimate the local model

and regenerate descriptive statistics. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The coefficient export used in this paper serves a specific empirical role: it is needed to recover the reported local coefficients and p-values. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The sample summary used in this paper serves a specific empirical role: it is needed to audit sample construction and raw grouped moments. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The fit summary used in this paper serves a specific empirical role: it is needed to check model diagnostics when a fit file is available. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, these files form the minimum executable evidence chain behind the paper. The theory comes from the source article, the estimable sample comes from the processed analysis panel, the regression narrative comes from the current coefficient export, and the sample audit comes from the local summary file. That is the basis on which the manuscript makes every empirical claim (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

Stage	NeededFor	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	Yes	Executable local dataset
Coefficient export	Recover the reported local coefficients and p-values	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	Yes	Executable local result file
Fit summary	Check model diagnostics when a fit file is available	Yes	Executable local result file

Verified evidence layers used to support the paper's calculations.

3.2 Sample Reconstruction

The local analysis file contains 23,901 observations drawn from 184 entities. The observed time span runs from 1986 to 2016. Selected sample moments are Outlier = 0.0064, Long Search Paths = 2.261, Scientific Reasoning = 6.772, Distant Recombination = 4.291, Recombination Degree = 0.555 (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

Statistic	Value
Observations	23,901
Unique Firms	184
Start Year	1986
End Year	2016
Mean Outlier	0.0064
Mean Long Search Paths	2.261
Mean Scientific Reasoning	6.772
Mean Distant Recombination	4.291
Mean Recombination Degree	0.555

Sample reconstruction from the current local dataset.

3.3 Variable Construction and Derivation

The construct Outlier innovation is operationalized locally as Top-tail impact indicator and appears in the executable dataset as high_top1_y. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the local yearly top-tail impact indicator.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Long search paths is operationalized locally as Log prior-art citations and appears in the executable dataset as long_search_paths. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Take log(1 + prior-art citations).. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation.

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same-year patents, prior stock and appears in the executable dataset as recombination_degree; total_patents_same_year; total_patents_until_prev_log. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reuse the patent-level controls already present in the merged analysis panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

Construct	Local measure	Derivation
Outlier innovation	Top-tail impact indicator	Use the local yearly top-tail impact indicator.
Long search paths	Log prior-art citations	Take log(1 + prior-art citations).
Scientific reasoning	Science-linked firm-year exposure	Use the processed firm-year science signal built from publication stock and NPL-related counts.
Distant recombination	Breadth of recombination	Carry forward the local recombination-breadth measure from the processed focal-patent panel.
Patent-level controls	Degree, same-year patents, prior stock	Reuse the patent-level controls already present in the merged analysis panel.

Variable mapping and derivation logic in the main text.

3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Kneeland, Schilling, and Aharonson (2020). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Outlier innovation, the manuscript uses Top-tail impact indicator under

the exact local implementation high_top1_y. The derivation step is: Use the local yearly top-tail impact indicator.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Long search paths, the manuscript uses Log prior-art citations under the exact local implementation long_search_paths. The derivation step is: Take log(1 + prior-art citations).. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Scientific reasoning, the manuscript uses Science-linked firm-year exposure under the exact local implementation scientific_reasoning. The derivation step is: Use the processed firm-year science signal built from publication stock and NPL-related counts.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Distant recombination, the manuscript uses Breadth of recombination under the exact local implementation distant_recombination. The derivation step is: Carry forward the local recombination-breadth measure from the processed focal-patent panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Patent-level controls, the manuscript uses Degree, same-year patents, prior stock under the exact local implementation recombination_degree; total_patents_same_year; total_patents_until_prev_log. The derivation step is: Reuse the patent-level controls already present in the merged analysis panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

The current dataset yields a richer descriptive picture than a single mean or coefficient can convey. Outlier indicator has mean 0.0064, standard deviation 0.0795, and an interquartile range from 0.00000 to 0.00000; Long search paths has mean 2.261, standard deviation 0.801, and an interquartile range from 1.792 to 2.639; Scientific reasoning has mean 6.772, standard deviation 2.130, and an interquartile range from 5.729 to 8.163; Distant recombination has mean 4.291, standard deviation 4.563, and an interquartile range from 2.000 to 5.000 (Kneeland et al., 2020; Jaffe, 1986;

Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The time-series view is equally informative. Patents changes from 646.0 in the first observed period to 534.0 in the last observed period; OutlierShare changes from 0.0046 in the first observed period to 0.0112 in the last observed period; MeanLongSearch changes from 1.807 in the first observed period to 2.208 in the last observed period . These temporal shifts help explain why a paper-comparable replication has to describe the support of the data before turning to regression output (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The regressor-quantile analysis shows that mean outcomes range from 0.0048 to 0.0115 across the support of the focal regressor, while the mean regressor itself ranges from 2.411 to 9.961. That pattern is useful because it shows what the formal model is trying to account for in the current dataset (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The correlation structure is reported in a compact form because it helps distinguish whether the main explanatory variables move in lockstep with the controls or whether the regression is drawing on independent variation. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

4.1 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. Outlier indicator has missing share 0.00%; Long search paths has missing share 0.00%; Scientific reasoning has missing share 0.00% . At the other end of the support, Outlier indicator remains close to complete with missing share 0.00%; Long search paths remains close to complete with missing share 0.00%; Scientific reasoning remains close to complete with missing share 0.00% . This matters because the credibility of the regression depends on whether the focal constructs survive sample cleaning with enough support to be interpreted substantively (Kneeland et al., 2020; Jaffe,

1986; Hall et al., 2005). Coverage across entities is also visible in the current panel rather than assumed. INTERNATIONAL BUSINESS MACHS COR contributes 3,200 observations; GENERAL ELECTRIC CO contributes 1,318 observations; EASTMAN KODAK CO contributes 1,017 observations; MINNESOTA MINING & MFG CO contributes 854 observations; DU PONT E I DE NEMOURS & CO contributes 830 observations . The purpose of reporting these concentrations is to show whether the estimated mechanism is being learned from a broad panel or from a narrow set of heavily observed entities (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). The support of the focal regressor is broad enough to permit a distributional reading in addition to the formal model. In the lowest observed quantile, the mean regressor is 2.411 and the mean outcome is 0.0048 across 13,716 observations. In the highest observed quantile, the corresponding values are 9.961, 0.0115, and 4,443 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). These data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because the paper is a rare strategy study that makes a strong process claim about how inventors reach territory far from what is already known. If the local panel were badly fragmented, dominated by missingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005).

Variable	Missing	MissingShare
Outlier indicator	0	0.00%
Long search paths	0	0.00%
Scientific reasoning	0	0.00%
Distant recombination	0	0.00%
Recombination degree	0	0.00%
Prior-art citations	0	0.00%
Science signal	0	0.00%
Same-year patents	0	0.00%
Prior cumulative stock	0	0.00%

Missingness profile for variables analyzed in the body of the paper.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Outlier indicator	23,901	0.0064	0.0795	0.0000	0.0000	0.0000	0	1
Long search paths	23,901	2.261	0.801	1.792	2.197	2.639	1	8
Scientific reasoning	23,901	6.772	2.130	5.729	7.149	8.163	0	10
Distant recombination	23,901	4.291	4.563	2.000	3.000	5.000	2	117
Recombination degree	23,901	0.555	0.174	0.444	0.560	0.691	0	1
Prior-art citations	23,901	15.07	50.39	5.000	8.000	13.00	2	2263
Science signal	23,901	6.772	2.130	5.729	7.149	8.163	0	10
Same-year patents	23,901	727.1	1,302	109.0	311.0	652.0	1	7321
Prior cumulative stock	23,901	7.500	2.217	6.324	7.684	8.893	0	11

Main descriptive statistics discussed in the body of the paper.

Period	Patents	OutlierShare	MeanLongSearch	MeanScientificReasoning	MeanDistantRecombination
1986	646	0.46%	1.807	6.156	2.865
1987	793	1.13%	1.863	6.285	2.914
1988	1,071	1.03%	1.914	6.229	3.135
1989	1,398	0.50%	1.925	6.217	3.084
1990	1,360	0.37%	1.974	6.370	3.117
1991	1,333	0.68%	1.974	6.436	3.151
1992	1,309	0.38%	2.055	6.486	3.329
1993	1,358	0.74%	2.117	6.861	3.465
1994	1,337	0.45%	2.182	6.866	3.572
1995	1,217	0.41%	2.223	6.983	3.844
1996	659	0.46%	2.310	6.880	3.838
1997	646	0.15%	2.358	6.781	4.245
1998	356	0.56%	2.265	6.949	3.924
1999	654	0.76%	2.295	6.895	3.810
2000	636	0.31%	2.286	6.781	4.022
2001	614	0.65%	2.389	6.943	4.414
2002	632	1.42%	2.394	6.915	4.384

Period	Patents	OutlierShare	MeanLongSearch	MeanScientificReasoning	MeanDistantRecombination
2003	522	0.77%	2.432	7.209	4.418
2004	324	0.00%	2.485	6.939	4.809
2005	284	1.06%	2.565	6.900	5.486
2006	336	0.89%	2.596	7.014	5.265
2007	246	0.00%	2.482	7.433	5.268
2008	535	0.19%	2.257	8.930	4.512
2009	206	1.46%	2.666	7.502	6.005
2010	511	0.00%	2.718	7.345	6.648
2011	651	0.46%	2.668	7.411	6.275
2012	1,068	0.19%	2.734	7.669	6.519
2013	868	0.69%	2.704	7.599	6.730
2014	725	1.93%	2.734	7.636	6.989
2015	1,072	1.03%	2.590	8.001	6.174
2016	534	1.12%	2.208	0.0000	4.629

Observed time patterns in the current dataset.

Quantile	N	MeanRegressor	MeanOutcome
Q1	13,716	2.411	0.0048
Q2	3,465	4.000	0.0058
Q3	2,277	5.000	0.0066
Q4	4,443	9.961	0.0115

Grouped summaries used to describe raw patterns in the current dataset.

Variable	Outlier indicator	Long search paths	Scientific reasoning	Distant recombination	Recombination degree	Prior-art citations	Science signal	Same-year patents	Prior cumulative stock
Outlier indicator	1.000	0.073	0.003	0.133	0.013	0.215	0.003	0	0.006
Long search paths	0.073	1.000	0.026	0.688	0.260	0.538	0.026	0	0.194
Scientific reasoning	0.003	0.026	1.000	0.023	0.037	0.024	1.000	0	0.398

Variable	Outlier indicator	Long search paths	Scientific reasoning	Distant recombination	Recombination degree	Prior-art citations	Science signal	Same-year patents	Prior cumulative stock
Distant recombination	0.133	0.688	0.023	1.000	0.466	0.822	0.023	0	0.149
Recombination degree	0.013	0.260	0.037	0.466	1.000	0.140	0.037	0	0.122
Prior-art citations	0.215	0.538	0.024	0.822	0.140	1.000	0.024	0	0.081

Compact correlation structure for the main variables.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is A patent-level fixed-effects count model for top-tail outlier status using the three process variables and patent-level controls. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest (Kneeland et al., 2020; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The first set of replication decisions concerns scope. The unit of analysis is Patent, the sample frame is Patent-level recombination panel with science-linked firm-year exposure merged in by firm and application year, and the empirical window is 1986 to 2016 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The second set of decisions concerns measurement and specification. The outcome is Top-tail outlier indicator from the local yearly patent-impact distribution, and the main explanatory design is Patent-level fixed-effects model relating outlier status to long search paths, scientific reasoning, and distant recombination. This is where the

replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Patent-level controls for degree, same-year patenting, prior cumulative stock, and available firm and year fixed effects.. The crucial exclusion or substitution rule is: The local rerun uses only patents with verified recombination measures and merges science exposure at the firm-year level when patent-specific scientific references are not directly recoverable.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. When the local data architecture differs, the paper marks the result as a calculated analogue rather than claiming byte-identical duplication (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

DecisionPoint	LocalSpecification
Research question	The source paper asks which search processes are associated with the origination of outlier innovation.
Unit of analysis	Patent
Sample frame	Patent-level recombination panel with science-linked firm-year exposure merged in by firm and application year.
Time window	1986 to 2016 in the local executable panel.
Outcome definition	Top-tail outlier indicator from the local yearly patent-impact distribution.

DecisionPoint	LocalSpecification
Main explanatory design	Patent-level fixed-effects model relating outlier status to long search paths, scientific reasoning, and distant recombination.
Controls and fixed effects	Patent-level controls for degree, same-year patenting, prior cumulative stock, and available firm and year fixed effects.
Exclusion or substitution logic	The local rerun uses only patents with verified recombination measures and merges science exposure at the firm-year level when patent-specific scientific references are not directly recoverable.

Replication decision register retained in the main text.

5.1 Why These Design Decisions Are Necessary

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Patent inside the sample frame Patent-level recombination panel with science-linked firm-year exposure merged in by firm and application year over 1986 to 2016 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Top-tail outlier indicator from the local yearly patent-impact distribution and the main explanatory design is Patent-level fixed-effects model relating outlier status to long search paths, scientific reasoning, and distant recombination. The local paper keeps these objects because they preserve the estimand's substantive

content while remaining executable from processed datasets already present in the workspace. Inference decisions: Patent-level controls for degree, same-year patenting, prior cumulative stock, and available firm and year fixed effects. The main exclusion or substitution rule is The local rerun uses only patents with verified recombination measures and merges science exposure at the firm-year level when patent-specific scientific references are not directly recoverable. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

$$E[\text{Outlier}_{pft} \mid X_{pft}] = \exp\big(\beta_1 \text{LongSearch}_p + \beta_2 \text{ScientificReasoning}_{ft} + \beta_3 \text{DistantRecombination}_p + \beta_4 X_{pft} + \alpha_f + \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is 0.0203 on Distant Recombination (p = 0.020). The local analogue finds positive and statistically meaningful coefficients on long search paths and distant recombination, but not on the firm-level science signal used to proxy scientific reasoning. Long search paths is estimated at 0.520 with p = ¡ .001. Log prior patent stock is estimated at -0.799 with p = 0.017. Distant recombination is estimated at 0.0203 with p = 0.020 (Kneeland et al., 2020; Hall et al., 2005). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Term	Estimate	Std. error	P-value
Long search paths	0.5199	0.1479	0.0004
Scientific reasoning	-0.0947	0.2188	0.665
Distant recombination	0.0203	0.0087	0.0197
Recombination degree	-0.3527	0.4690	0.452
Same-year patents	-3.14e-05	0.0002	0.879
Log prior patent stock	-0.7989	0.3339	0.0167

Key coefficient table used in the main text.

6.2 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Long search paths is estimated at 0.520 with a p-value of ¡ .001; Log prior patent stock is estimated at -0.799 with a p-value of 0.017; Distant recombination is estimated at 0.0203 with a p-value of 0.020; Recombination degree is estimated at -0.353 with a p-value of 0.45 (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Within the focal specification, the sharpest local signal comes from Long search paths, which is estimated at 0.520 with p = ¡ .001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The scale of the underlying variables also matters for interpretation. The outcome Outlier indicator has mean 0.0064 and standard deviation 0.0795 in the local panel, while the main regressor Distant recombination has mean 4.291 and standard deviation 4.563 (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 0.0048 to 0.0115 while the mean regressor ranges from 2.411 to 9.961. This does not substitute for the model, but it shows how much raw variation the specification must explain (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local analogue finds positive and statistically meaningful coefficients on long search paths and distant recombination, but not on the firm-level science signal used to proxy scientific reasoning. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

6.3 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within baseline, the leading coefficient family reads as follows: Long search paths = 0.520 (p = ¡ .001); Log prior patent stock = -0.799 (p = 0.017); Distant recombination = 0.0203 (p = 0.020); Recombination degree = -0.353 (p = 0.45); Scientific reasoning = -0.0947 (p = 0.67). The largest positive estimate in this specification is Long search paths at 0.520, while the largest negative estimate is Log prior patent stock at -0.799. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This fuller coefficient reading is especially important for Kneeland, Schilling, and Aharonson (2020) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text (Kneeland et al., 2020; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

6.4 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Baseline, the most informative focal term is Long search paths with estimate 0.520 and p = ¡ .001 (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Baseline specification contributes its own reading to the replication. Long search paths is estimated at 0.520 with p = ¡ .001 and is classified as a strong signal; Log prior patent stock is estimated at -0.799 with p = 0.017 and is classified as a moderate signal; Distant recombination is estimated at 0.0203 with p = 0.020 and is classified as a moderate signal . This matters because

the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This model comparison matters for Kneeland, Schilling, and Aharonson (2020) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

Term	Model	Estimate	PValue	Signal	Reading
Long search paths	Baseline	0.5199	¡ .001	Strong	Positive estimate with p = ¡ .001.
Scientific reasoning	Baseline	-0.0947	0.67	Weak	Negative estimate with p = 0.67.
Distant recombination	Baseline	0.0203	0.020	Moderate	Positive estimate with p = 0.020.
Recombination degree	Baseline	-0.3527	0.45	Weak	Negative estimate with p = 0.45.
Same-year patents	Baseline	-3.14e-05	0.88	Weak	Negative estimate with p = 0.88.
Log prior patent stock	Baseline	-0.7989	0.017	Moderate	Negative estimate with p = 0.017.

Model-by-model interpretation table drawn from the current coefficient exports.

6.5 Diagnostic Evidence

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a sample and likelihood structure that are transparent enough to justify interpretation (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). For outlier_processes, the current fit export reports N = 16,820, pseudo R-squared = 0.135 (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005). These diagnostics do not, by themselves, validate the source paper's theory. They do, however, show that the local data analysis behind Kneeland, Schilling, and Aharonson (2020) is being carried out on a clearly described estimating sample with observable fit statistics (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005).

model	n	logLik	AIC	BIC	pseudo_r2
outlier_processes	16,820	-750.7	1,643	2,192	0.135

Model fit statistics discussed in the main results section.

Term	Estimate	Std. error	Statistic	P-value
Long search paths	0.5199	0.1479	3.515	0.0004
Scientific reasoning	-0.0947	0.2188	-0.433	0.665
Distant recombination	0.0203	0.0087	2.333	0.0197
Recombination degree	-0.3527	0.4690	-0.752	0.452
Same-year patents	-3.14e-05	0.0002	-0.153	0.879
Log prior patent stock	-0.7989	0.3339	-2.392	0.0167

Detailed coefficient table from the current local estimation output.

7 Comparison to the Published Study

7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central empirical claim, the main measurement layer, and the effective sample definition are set alongside the current local dataset result so the reader can see where agreement is substantive and where it is conditional on different data architecture. (Kneeland et al., 2020). For Kneeland, Schilling, and Aharonson (2020), the local headline result is Distant Recombination = 0.0203 with p = 0.020. The value of presenting that estimate directly against the original paper's reported finding is that it shows whether the mechanism survives a different but transparent processed dataset rather than asking the reader to infer comparability from prose alone.

Aspect	OriginalPaper	LocalDataset	Reading
Main empirical result	The published paper argues that outlier innovations are more likely to emerge from long search paths, scientific reasoning, and distant recombination.	Distant Recombination = 0.0203 with p = 0.020.	Two of the three source-paper processes carry through more clearly than the third in the local dataset. Long search paths and distant recombination are positive and significant, whereas the science-linked proxy for scientific reasoning is not.

Aspect	OriginalPaper	LocalDataset	Reading
Measurement and data construction	The published design identifies outlier patents in technological space and studies the search processes behind them.	The local analogue maps the three proposed processes onto a processed patent panel: long search paths are proxied by prior-art search depth, scientific reasoning by science-linked firm-year exposure, and distant recombination by recombination breadth.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the same substantive target.
Sample and unit of analysis	The published study is a patent-level archival analysis supplemented by inventor interviews.	Patent-level recombination panel with science-linked firm-year exposure merged in by firm and application year. Local sample = 23,901 observations across 184 entities.	This row clarifies whether agreement or disagreement with the original paper could reflect sample translation rather than theory alone.

Comparison of the original paper's results and design to the current local dataset analysis.

7.2 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published design identifies outlier patents in technological space and studies the search processes behind them By contrast, the local paper relies on the following executable design: The local

analogue maps the three proposed processes onto a processed patent panel: long search paths are proxied by prior-art search depth, scientific reasoning by science-linked firm-year exposure, and distant recombination by recombination breadth These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The same caution applies to the sample itself. The published design is described as follows: The published study is a patent-level archival analysis supplemented by inventor interviews The local sample instead works from Patent-level recombination panel with science-linked firm-year exposure merged in by firm and application year and yields 23,901 observations across 184 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. Two of the three source-paper processes carry through more clearly than the third in the local dataset. Long search paths and distant recombination are positive and significant, whereas the science-linked proxy for scientific reasoning is not. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The comparison to the published paper is intentionally narrow. Two of the three source-paper processes carry through more clearly than the third in the local dataset. Long search paths and distant recombination are positive and significant, whereas the science-linked proxy for scientific reasoning is not. The manuscript therefore evaluates concordance at the level of mechanism, sign

pattern, and construct behavior before drawing any conclusion about exact numerical replication (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

8 Discussion

Two of the three source-paper processes carry through more clearly than the third in the local dataset. Long search paths and distant recombination are positive and significant, whereas the science-linked proxy for scientific reasoning is not (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (Kneeland et al., 2020; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The first limitation concerns measurement. The source article identifies outliers in technological space, whereas the local rerun uses top-tail inventive impact as the executable outlier proxy (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The second limitation concerns sample translation. Scientific reasoning is proxied with firm-level science exposure rather than inventor interviews or patent-specific scientific narratives (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The third limitation concerns interpretation. The local paper is therefore a calculated analogue of the process logic, not a literal reconstruction of the original outlier-identification algorithm

(Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005).

9 Conclusion

The paper's contribution is a transparent processed-dataset analogue that preserves the three-process logic while making the outlier and science proxies explicit (Kneeland et al., 2020; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (Kneeland et al., 2020; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value	RawStatistic
Observations	23,901	rows
Unique Firms	184	unique_firms
Start Year	1986	year_min
End Year	2016	year_max
Mean Outlier	0.0064	mean_outlier
Mean Long Search Paths	2.261	mean_long_search_paths
Mean Scientific Reasoning	6.772	mean_scientific_reasoning
Mean Distant Recombination	4.291	mean_distant_recombination
Mean Recombination Degree	0.555	mean_recombination_degree

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Outlier indicator	23,901	0.0064	0.0795	0.0000	0.0000	0.0000	0	1
Long search paths	23,901	2.261	0.801	1.792	2.197	2.639	1	8
Scientific reasoning	23,901	6.772	2.130	5.729	7.149	8.163	0	10
Distant recombination	23,901	4.291	4.563	2.000	3.000	5.000	2	117
Recombination degree	23,901	0.555	0.174	0.444	0.560	0.691	0	1
Prior-art citations	23,901	15.07	50.39	5.000	8.000	13.00	2	2263
Science signal	23,901	6.772	2.130	5.729	7.149	8.163	0	10
Same-year patents	23,901	727.1	1,302	109.0	311.0	652.0	1	7321
Prior cumulative stock	23,901	7.500	2.217	6.324	7.684	8.893	0	11

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Outlier indicator	Long search paths	Scientific reasoning	Distant recombination	Recombination degree	Prior-art citations	Science signal	Same-year patents	Prior cumulative stock
Outlier indicator	1.000	0.073	0.003	0.133	0.013	0.215	0.003	0	0.006
Long search paths	0.073	1.000	0.026	0.688	0.260	0.538	0.026	0	0.194
Scientific reasoning	0.003	0.026	1.000	0.023	0.037	0.024	1.000	0	0.398
Distant recombination	0.133	0.688	0.023	1.000	0.466	0.822	0.023	0	0.149
Recombination degree	0.013	0.260	0.037	0.466	1.000	0.140	0.037	0	0.122
Prior-art citations	0.215	0.538	0.024	0.822	0.140	1.000	0.024	0	0.081
Science signal	0.003	0.026	1.000	0.023	0.037	0.024	1.000	0	0.398
Same-year patents	-0.011	0.016	0.463	0.011	0.065	-0.016	0.463	1	0.520
Prior cumulative stock	0.006	0.194	0.398	0.149	0.122	0.081	0.398	1	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	Patents	OutlierShare	MeanLongSearch	MeanScientificReasoning	MeanDistantRecombination
1986	646	0.46%	1.807	6.156	2.865
1987	793	1.13%	1.863	6.285	2.914
1988	1,071	1.03%	1.914	6.229	3.135
1989	1,398	0.50%	1.925	6.217	3.084
1990	1,360	0.37%	1.974	6.370	3.117
1991	1,333	0.68%	1.974	6.436	3.151
1992	1,309	0.38%	2.055	6.486	3.329
1993	1,358	0.74%	2.117	6.861	3.465
1994	1,337	0.45%	2.182	6.866	3.572
1995	1,217	0.41%	2.223	6.983	3.844
1996	659	0.46%	2.310	6.880	3.838
1997	646	0.15%	2.358	6.781	4.245
1998	356	0.56%	2.265	6.949	3.924
1999	654	0.76%	2.295	6.895	3.810
2000	636	0.31%	2.286	6.781	4.022
2001	614	0.65%	2.389	6.943	4.414
2002	632	1.42%	2.394	6.915	4.384
2003	522	0.77%	2.432	7.209	4.418
2004	324	0.00%	2.485	6.939	4.809
2005	284	1.06%	2.565	6.900	5.486
2006	336	0.89%	2.596	7.014	5.265
2007	246	0.00%	2.482	7.433	5.268
2008	535	0.19%	2.257	8.930	4.512
2009	206	1.46%	2.666	7.502	6.005
2010	511	0.00%	2.718	7.345	6.648
2011	651	0.46%	2.668	7.411	6.275
2012	1,068	0.19%	2.734	7.669	6.519
2013	868	0.69%	2.704	7.599	6.730

Period	Patents	OutlierShare	MeanLongSearch	MeanScientificReasoning	MeanDistantRecombination
2014	725	1.93%	2.734	7.636	6.989
2015	1,072	1.03%	2.590	8.001	6.174
2016	534	1.12%	2.208	0.0000	4.629

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
Q1	13,716	2.411	0.0048
Q2	3,465	4.000	0.0058
Q3	2,277	5.000	0.0066
Q4	4,443	9.961	0.0115

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value
long_search_paths	0.5199	0.1479	3.515	¡ .001
scientific_reasoning	-0.0947	0.2188	-0.433	0.67
distant_recombination	0.0203	0.0087	2.333	0.020
recombination_degree	-0.3527	0.4690	-0.752	0.45
total_patents_same_year	-3.14e-05	0.0002	-0.153	0.88
total_patents_until_prev_log	-0.7989	0.3339	-2.392	0.017

Full coefficient dump exported from the local estimation script.

G Appendix G: Coverage Tables

Variable	Missing	MissingShare
Outlier indicator	0	0.00%
Long search paths	0	0.00%
Scientific reasoning	0	0.00%
Distant recombination	0	0.00%
Recombination degree	0	0.00%
Prior-art citations	0	0.00%
Science signal	0	0.00%
Same-year patents	0	0.00%
Prior cumulative stock	0	0.00%

Missingness audit for the variables summarized in the manuscript.

Entity	Label	Observations	FirstPeriod	LastPeriod
6066	INTERNATIONAL BUSINESS MACHS COR	3,200	1986	2016
5047	GENERAL ELECTRIC CO	1,318	1986	2016
4194	EASTMAN KODAK CO	1,017	1986	2010
7435	MINNESOTA MINING & MFG CO	854	1986	2016
4087	DU PONT E I DE NEMOURS & CO	830	1986	2016
4839	FORD MOTOR CO DEL	815	1986	2016
7585	MOTOROLA INC	774	1986	2016
10499	TEXAS INSTRUMENTS INC	755	1986	2016

Entity	Label	Observations	FirstPeriod	LastPeriod
5073	GENERAL MOTORS CORP	671	1986	2008
4060	DOW CHEMICAL CO	574	1986	2016
1300	ALLIED SIGNAL INC	551	1986	2016
8762	PROCTER & GAMBLE CO	524	1986	2016
1690	APPLE COMPUTER INC	452	1990	2016
1581	AMERICAN TELEPHONE & TELEG CO	421	1986	2004
10983	UNITED TECHNOLOGIES CORP	403	1986	2016
12141	MICROSOFT CORP	398	1995	2016
4503	EXXON CORP	344	1986	2016
6266	JOHNSON & JOHNSON	340	1986	2016
2285	BOEING CO	334	1986	2016
7475	MOBIL CORP	300	1986	1998

Top entities by local observation count.

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